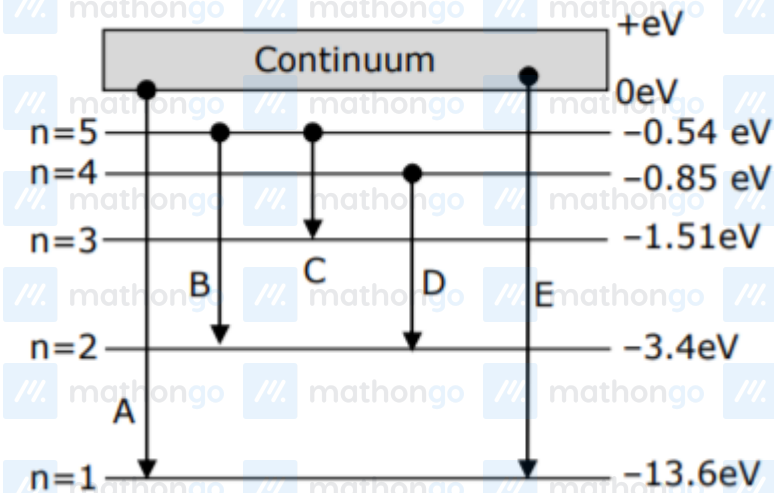


Q1: 24 Feb (Shift 1) - Single Correct

In the given figure, the energy levels of hydrogen atom have been shown along with some transitions marked A, B, C, D and E. The transitions A, B and C respectively represents -



- (1) The series limit of Lyman series, third member of Balmer series and second member of Paschen series
- (2) The first member of the Lyman series, third member of Balmer series and second member of Paschen series
- (3) The ionization potential of hydrogen, second member of Balmer series and third member of Paschen series
- (4) The series limit of Lyman series, second member of Balmer series and second member of Paschen series.

Q2: 24 Feb (Shift 2) - Single Correct

According to Bohr atom model, in which of the following transitions will the frequency be maximum?

- (1) $n = 2$ to $n = 1$
- (2) $n = 4$ to $n = 3$
- (3) $n = 5$ to $n = 4$
- (4) $n = 3$ to $n = 2$

Q3: 24 Feb (Shift 2) - Single Correct

The de Broglie wavelength of a proton and α -particle are equal. The ratio of their velocities is :

- (1) 4: 2

Questions with Answer Keys

MathonGo

(2) 4: 1

(3) 1: 4

(4) 4: 3

Q4: 25 Feb (Shift 1) - Single Correct

An α particle and a proton are accelerated from rest by a potential difference of 200 V. After this, their de Broglie wavelengths are λ_α and λ_p respectively. The ratio $\frac{\lambda_p}{\lambda_\alpha}$ is :

(1) 8

(2) 2.8

(3) 3.8

(4) 7.8

Q5: 25 Feb (Shift 2) - Single Correct

The wavelength of the photon emitted by a hydrogen atom when an electron makes a transition from $n = 2$ to $n = 1$ state is:

(1) 194.8nm

(2) 490.7nm

(3) 913.3nm

(4) 121.8nm

Q6: 25 Feb (Shift 2) - Single Correct

The stopping potential for electrons emitted from a photosensitive surface illuminated by light of wavelength 491nm is 0.710 V. When the incident wavelength is changed to a new value, the stopping potential is 1.43 V.

The new wavelength is:

(1) 400nm

(2) 382nm

(3) 309nm

Questions with Answer Keys

MathonGo

(4) 329nm

Q7: 25 Feb (Shift 2) - Single Correct

An electron of mass m_e and a proton of mass $m_p = 1836m_e$ are moving with the same speed. The ratio of their de Broglie wavelength $\frac{\lambda_{\text{electron}}}{\lambda_{\text{proton}}}$ will be:

(1) 918

(2) 1836

(3) $\frac{1}{1836}$

(4) 1

Q8: 26 Feb (Shift 1) - Single Correct

If λ_1 and λ_2 are the wavelengths of the third member of Lyman and first member of the Paschen series respectively, then the value of $\lambda_1 : \lambda_2$ is :

(1) 1: 3

(2) 1: 9

(3) 7:135

(4) 7:108

Q9: 26 Feb (Shift 2) - Single Correct

The recoil speed of a hydrogen atom after it emits a photon in going from $n = 5$ state to $n = 1$ state will be:

(1) 4.17 m/s

(2) 4.34 m/s

(3) 219 m/s

(4) 3.25 m/s

Q10: 26 Feb (Shift 2) - Numerical

Two stream of photons, possessing energies equal to twice and ten times the work function of metal are incident on the metal surface successively. The value of ratio of maximum velocities of the photoelectrons emitted in the two respective cases is $x : y$. The value of x is

Answer Key**Q1 (1)****Q2 (1)****Q3 (2)****Q4 (2)****Q5 (4)****Q6 (2)****Q7 (2)****Q8 (3)****Q9 (1)****Q10 (1)**

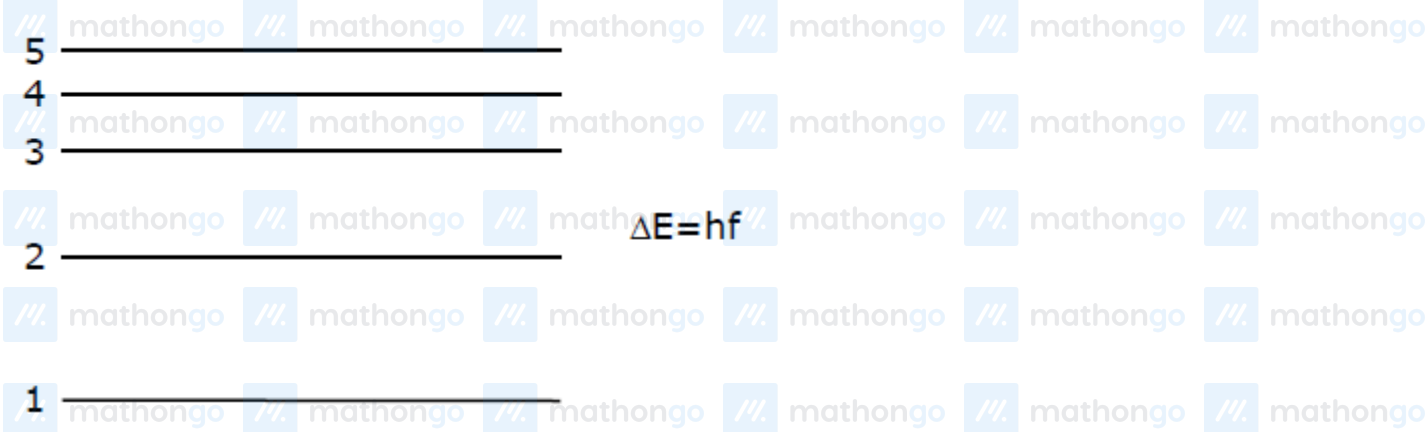
Hints and Solutions

MathonGo

Q1 (1)

A $\rightarrow$ series limit of Lyman.B $\rightarrow$ 3rd member of Balmer series.C $\rightarrow$ 2nd member of Paschen series.

Q2 (1)

f is more for transition from $n = 2$ to $n = 1$.

Q3 (2)

From De-broglie's wavelength: $\lambda = \frac{h}{mv}$ Given $\lambda_p = \lambda_\alpha$

$$v_\alpha \frac{1}{m} = \frac{v_p}{\frac{m_\alpha}{m_p}} = \frac{4 m_p}{m_p} = \frac{4}{1}$$

Q4 (2)

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqv}}$$

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}} = \sqrt{\frac{4 \times 2}{1 \times 1}}$$

$$= 2\sqrt{2} = 2.8$$

Q5 (4)

Hints and Solutions

MathonGo

$$\Delta E = 10.2 \text{ eV}$$

$$\frac{hc}{\lambda} = 10.2 \text{ eV}$$

$$\lambda = \frac{hc}{(10.2)e}$$

$$= \frac{12400}{10.2} \text{ \AA}$$

$$= 121.56 \text{ nm}$$

$$\simeq 121.8 \text{ nm}$$

Q6 (2)

From the photoelectric effect equation

$$\frac{hc}{\lambda} = \phi + ev_s$$

$$\text{so } ev_{s_1} = \frac{hc}{\lambda_1} - \phi \dots\dots$$

$$ev_{s_2} = \frac{hc}{\lambda_2} - \phi \dots\dots \text{(ii)}$$

$$\text{Subtract equation (i) from equation (ii) } ev_{s_1} - ev_{s_2} = \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2}$$

$$v_{s_1} - v_{s_2} = \frac{hc}{e} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)$$

$$(0.710 - 1.43) = 1240 \left(\frac{1}{491} - \frac{1}{\lambda_2} \right)$$

$$\frac{-0.72}{1240} = \frac{1}{491} - \frac{1}{\lambda_2}$$

$$\frac{1}{\lambda_2} = \frac{1}{491} + \frac{0.72}{1240}$$

$$\frac{1}{\lambda_2} = 0.00203 + 0.00058$$

$$\frac{1}{\lambda_2} = 0.00261$$

$$\lambda_2 = 383.14$$

$$\lambda_2 \simeq 382 \text{ nm}$$

Q7 (2)

Given mass of electron = m_e

Mass of proton = m_p

$\therefore$ given $m_p = 1836 m_e$

From de-Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Hints and Solutions

MathonGo

$$\frac{\lambda_e}{\lambda_p} = \frac{m_p}{m_e}$$

$$= \frac{1836 m_e}{m_e}$$

$$\frac{\lambda_e}{\lambda_p} = 1836$$

Q8 (3)

For Lyman series

$$n_1 = 1, \quad n_2 = 4$$

$$\frac{1}{\lambda_1} = R_z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda_1} = R_z^2 \left(\frac{1}{1^2} - \frac{1}{4^2} \right)$$

$$\frac{1}{\lambda_1} = \frac{15R_z^2}{16}$$

$$\lambda_1 = \frac{16}{15R_z^2}$$

For paschen series

$$n_1 = 3, \quad n_2 = 4$$

$$\frac{1}{\lambda_2} = R_z^2 \left(\frac{1}{3^2} - \frac{1}{4^2} \right)$$

$$\frac{1}{\lambda_2} = R_z^2 \left(\frac{16-9}{9 \times 16} \right)$$

$$\frac{1}{\lambda_2} = R_z^2 \left(\frac{7}{9 \times 16} \right)$$

$$\lambda_2 = \frac{9 \times 16}{7R_z^2}$$

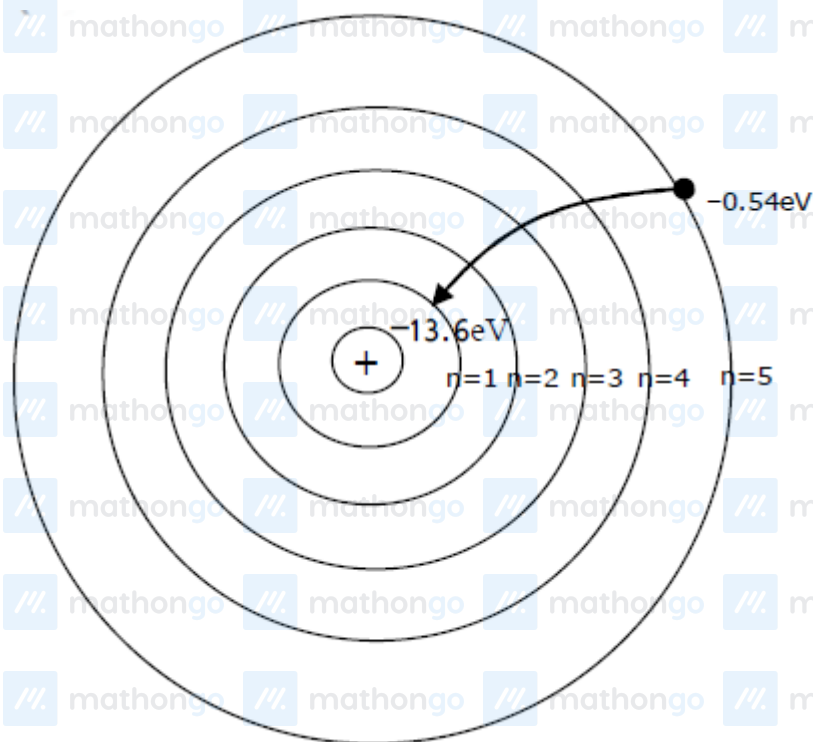
So,

$$\frac{\lambda_1}{\lambda_2} = \frac{\frac{16}{15R_z^2}}{\frac{9 \times 16}{7R_z^2}}$$

$$= \frac{16 \times 7}{15 \times 9 \times 16}$$

$$= \frac{7}{135}$$

Q9 (1)



$$\text{momentum } (P) = \frac{\Delta E}{C} = \frac{(13.6 - 0.54) \text{ eV}}{3 \times 10^8}$$

$$mv = \frac{(13.06) \times 1.6 \times 10^{-19}}{3 \times 10^8}$$

$$v = \frac{(13.06) \times 1.6 \times 10^{-19}}{3 \times 10^8 \times 1.67 \times 10^{-27}} \Rightarrow 4.17 \text{ m/sec}$$

Q10 (1)

For photoelectric effect, $E_{\text{max}} = E - \phi$

$$E_1 = 2\phi_r \quad k_1 = \phi$$

$$E_2 = 10\phi_r \quad k_2 = 9\phi$$

$$\therefore V \propto \sqrt{k} \quad \left(k = \frac{1}{2}mv^2\right)$$

$$\frac{v_1}{v_2} = \sqrt{\frac{1}{9}} \Rightarrow \frac{1}{3} = \frac{x}{y}$$

$$x = 1$$